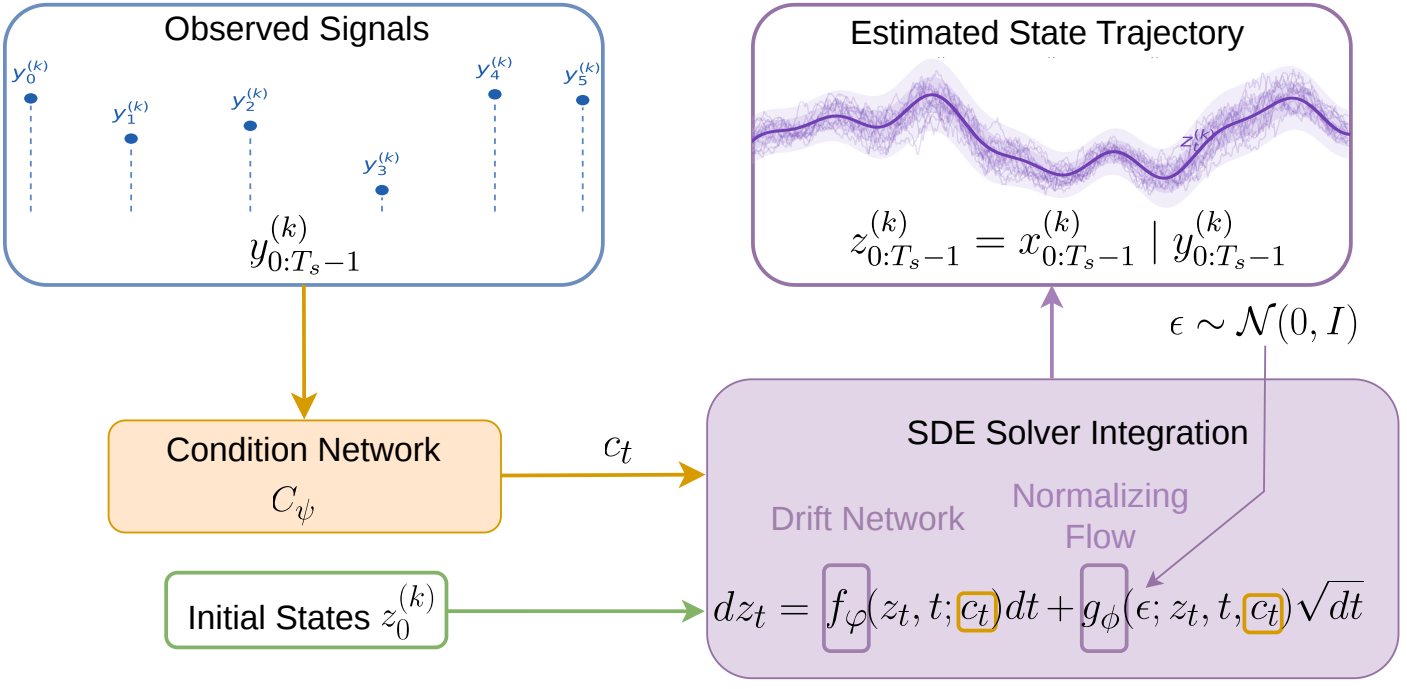

Observed Signals
$y_0^{(k)}$
$y_1^{(k)}$
$y_2^{(k)}$
$y_3^{(k)}$
$y_4^{(k)}$
$y_5^{(k)}$
$y^{(k)}_{0:T_s-1}$
Estimated State Trajectory
$z_t^{(k)}$
$z^{(k)}_{0:T_s-1} = x^{(k)}_{0:T_s-1} \mid y^{(k)}_{0:T_s-1}$
$\epsilon \sim \mathcal{N}(0, I)$
Condition Network
C_ψ
c_t
SDE Solver Integration
Drift Network
Normalizing Flow
$dz_t = f_\varphi(z_t, t; c_t)dt + g_\phi(\epsilon; z_t, t, c_t)\sqrt{dt}$
Initial States $z_0^{(k)}$